PASSAGE: A Multi-Resolution Pathfinding Dataset mixing Complex Obstacles and Elevation-based tiles towards Certified Machine Learning Systems

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Abstract

We present PASSAGE, a large-scale, multi-resolution pathfinding benchmark derived from the JAXA ALOS AW3D30 Digital Surface Model. Unlike existing grid-based benchmarks that rely on binary obstacles and fixed resolutions, PASSAGE provides continuous elevation-derived cost maps, procedurally generated superellipse obstacles with configurable shape distributions, and ground-truth paths computed by multiple solver backends. The dataset spans seven resolution levels (64×64 to 4096×4096 pixels) with 141,300 samples in repository defaults, organized into stratified train/calibration/validation/test splits (0.90/0.02/0.04/0.04). We provide five physically-motivated cost models (elevation, distance, slope-uphill, energy, absolute-elevation) and three solver backends (NetworkX, pathfinding, NumPy-grid) supporting A*, Dijkstra, and bidirectional variants. Designed with aerospace certification standards (ARP6983, ED-324) in mind, PASSAGE offers deterministic generation, complete provenance metadata (including per-sample timing breakdowns), and a dedicated calibration split for certification-oriented experiments. The toolkit includes direct Hugging Face Hub export with zstd-compressed shards and configurable chunk sizes. We release the full generation pipeline, configuration utilities, analysis notebooks, and baseline experiments to enable reproducible research in supervised surrogate modeling, reinforcement learning, and graph neural networks for pathfinding. The dataset and code are available at https://huggingface.co/datasets/passage.

1 Introduction

1.1 Context and Motivation

The proliferation of learning-based approaches for navigation and pathfinding—including supervised surrogate models, reinforcement learning (RL) agents, and graph neural networks (GNNs)—has created substantial demand for standardized benchmark datasets. Such benchmarks are essential for fair comparison of algorithms, reproducible research, and systematic evaluation of generalization capabilities.

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However, existing pathfinding benchmarks suffer from several critical limitations. Grid-based benchmarks from the Moving AI Lab Sturtevant (2012) provide maps derived from video games (Dragon Age, StarCraft) but feature only binary obstacles and fixed resolutions, lacking the continuous cost gradients found in real terrain. Random grid environments offer scalability but sacrifice terrain realism. Continuous navigation datasets from robotics and autonomous driving (*e.g.*, nuScenes, Waymo) incorporate sensor modalities and dynamics but are unsuitable for evaluating discrete grid-based algorithms.

Simultaneously, safety-critical domains such as aerospace, autonomous vehicles, and robotics increasingly require *certifiable* AI systems operating within well-defined Operational Design Domains (ODD). Emerging standards like SAE ARP6983 SAE International (2024) and EUROCAE ED-324 EUROCAE WG-114 (2024) mandate reproducible test conditions, bounded performance metrics, and evidence of data quality—requirements that existing pathfinding datasets do not adequately address.

1.2 Problem Statement

We identify the following gaps in current pathfinding datasets:

- **Limited resolution variety**: Most benchmarks offer single or few fixed resolutions, preventing studies of computational scalability and resolution-dependent generalization.
- **Binary obstacles only**: Existing datasets lack continuous terrain-based cost models that capture elevation, slope, and energy expenditure.
- **Insufficient sample counts**: Many benchmarks provide too few samples for modern deep learning architectures requiring thousands of training examples.
- **No certification alignment**: Current datasets lack the provenance metadata, deterministic generation, and split structures needed for certification workflows.
- **Single solver ground truth**: Most datasets provide paths from a single algorithm, preventing studies of solver agreement and algorithm selection.

1.3 Contributions

We address these gaps with the following contributions:

1. **PASSAGE Dataset**: A large-scale, multi-resolution pathfinding benchmark with 141,300 samples (repository defaults) across seven resolutions (64×64 to 4096×4096), derived from real-world elevation data (JAXA ALOS AW3D30).
2. **Procedural Obstacle Generation**: Configurable superellipse-based obstacles (Lamé curves) with flexible shape distributions (log-uniform, uniform, beta, exponential), size ratios, rotation angles, and coverage constraints up to 30%.
3. **Multi-Backend Solver Framework**: Three solver backends (NetworkX, pathfinding, NumPy-grid) implementing A*, Dijkstra, bidirectional variants, and Bellman-Ford, with five physically-motivated cost models.
4. **Certification-Oriented Design**: Bounded ODD definition, deterministic generation with configurable seeds, complete provenance metadata per sample (including timing traces), and a dedicated calibration split (2%) for conformal prediction and uncertainty quantification experiments.
5. **Open-Source Toolkit**: Complete pipeline from tile download through Hugging Face Hub export, with zstd-compressed shards and configurable per-resolution chunk sizes for efficient storage and streaming.

1.4 Paper Organization

Section 2 reviews related work on pathfinding datasets, learning-based approaches, and certification standards. Section 3 presents the dataset design, including data sources, obstacle generation, and cost models. Section 4 describes the generation pipeline and export mechanisms. Section 5 details solver implementations and evaluation metrics. Section 6 presents experimental validation including solver benchmarks and baseline

learning models. Section 7 proposes research directions enabled by PASSAGE. Section 10 concludes with future work.

2 Related Work

2.1 Existing Pathfinding Datasets

Grid-Based Benchmarks. The Moving AI Lab benchmarks Sturtevant (2012) remain the most widely used pathfinding testbeds, providing maps from Dragon Age, StarCraft, and other games. While valuable for algorithm comparison, these maps feature binary obstacles (passable/impassable) and fixed resolutions, limiting their applicability to terrain-aware planning. Random grid generators offer scalability but lack geographic realism.

Continuous Environment Datasets. Robot navigation datasets Mirowski et al. (2017) and autonomous driving benchmarks (nuScenes Caesar et al. (2020), Waymo Sun et al. (2020)) operate in continuous state spaces with rich sensor modalities. However, their focus on perception and dynamics makes them unsuitable for evaluating discrete grid-based pathfinding algorithms.

Elevation-Based Data. Digital Elevation Models (DEMs) from sources like SRTM, ASTER, and ALOS are available for GIS applications but lack standardized pathfinding ground truth, train/test splits, and obstacle annotations suitable for ML evaluation.

2.2 Learning-Based Pathfinding

Supervised Surrogate Modeling. Convolutional neural networks have been applied to path prediction, with U-Net architectures Ronneberger et al. (2015) producing path segmentation masks. Encoder-decoder models estimate cost-to-go functions, while attention mechanisms capture long-range dependencies. However, these approaches typically assume fixed input resolutions and struggle with generalization across terrain types.

Value Iteration Networks. Tamar *et al.* Tamar et al. (2016) introduced Value Iteration Networks (VIN), embedding differentiable planning modules within neural networks. Extensions include GVIN for graphs and hierarchical variants for multi-scale planning.

Neural A* and Learned Heuristics. Yonetani *et al.* Yonetani et al. (2021) proposed Neural A*, learning cost maps that guide classical A* search. This hybrid approach combines neural flexibility with algorithmic guarantees.

Graph Neural Networks. GNNs naturally represent grid connectivity and have shown promise for combinatorial optimization Cappart et al. (2023). Message Passing Neural Networks (MPNNs) propagate information along edges, learning path-relevant features.

Reinforcement Learning. Deep Q-Networks (DQN), policy gradient methods (A2C, PPO), and hierarchical RL have been applied to navigation Mirowski et al. (2017). Challenges include sample efficiency, reward shaping, and generalization to unseen environments.

2.3 Certification Standards for AI Systems

Aerospace Standards. SAE ARP6983 SAE International (2024) provides process guidelines for developing and certifying AI-based aeronautical products. EUROCAE ED-324 (developed by WG-114) EUROCAE WG-114 (2024) offers specific guidance for machine learning in aviation. Key requirements include:

- Defined Operational Design Domain (ODD) with explicit boundaries
- Data quality assurance and representativeness evidence
- Performance monitoring with bounded metrics
- Traceability and provenance documentation

Challenges for ML-Based Pathfinding Certification. Certifying neural pathfinders requires addressing formal verification of network properties, worst-case performance guarantees, distribution shift detection, and explainability. Conformal prediction Vovk et al. (2005) offers promising uncertainty quantification with coverage guarantees.

3 Dataset Design

3.1 Data Source: JAXA ALOS AW3D30

PASSAGE uses the Advanced Land Observing Satellite (ALOS) World 3D-30m Digital Surface Model (AW3D30) provided by the Japan Aerospace Exploration Agency (JAXA) Tadono et al. (2014):

- **Resolution**: 30-meter ground sampling distance (1 arcsecond)
- **Coverage**: Near-global ($\pm 60^\circ$ latitude)
- **Format**: GeoTIFF tiles, 3600×3600 pixels per $1^\circ \times 1^\circ$ tile
- **Elevation units**: Meters above EGM96 geoid
- **Accuracy**: $\sim$5m vertical RMSE

We selected AW3D30 for its global coverage, consistent quality, and free availability for research purposes.

3.2 Operational Design Domain (ODD)

Following certification guidelines, we explicitly define the ODD:

Table 1. Operational Design Domain specification for PASSAGE.

Parameter	Bounds
Geographic latitude	$[-60^\circ, +60^\circ]$
Geographic longitude	$[-180^\circ, +180^\circ]$
Elevation range	[calibrated global min, max]
Grid resolutions	$\{64, 128, 256, 512, 1024, 2048, 4096\}$
Obstacle coverage	$[0\%, 30\%]$
Superellipse exponent n	$[1.0, 80.0]$
Connectivity	4-connected or 8-connected

3.3 Multi-Resolution Support

We generate samples at seven resolution levels (Table 2), enabling studies of:

- **Computational scalability**: How solver time grows with grid size
- **Resolution-dependent performance**: Whether models trained at one resolution transfer to others
- **Hierarchical approaches**: Multi-scale planning strategies

Sample counts are intentionally skewed toward smaller resolutions where training is computationally feasible, while larger resolutions provide challenging test cases.

3.4 Dataset Splits

We provide stratified splits for reproducible evaluation:

The **calibration split** is specifically designed for certification-oriented experiments including conformal prediction, uncertainty calibration, and threshold tuning without contaminating validation or test sets.

Table 2. Resolution levels and repository-default sample counts in PASSAGE.

Resolution	Samples	Grid Cells	Physical Extent	Complexity
64×64	100,000	4,096	1.9×1.9 km	Lightweight
128×128	25,000	16,384	3.8×3.8 km	Small
256×256	10,000	65,536	7.7×7.7 km	Medium
512×512	5,000	262,144	15.4×15.4 km	Large
1024×1024	1,000	1,048,576	30.7×30.7 km	Heavy
2048×2048	200	4,194,304	61.4×61.4 km	Very Heavy
4096×4096	100	16,777,216	122.9×122.9 km	Extreme
Total	**141,300**	—	—	—

Table 3. Dataset split ratios (repository defaults).

Split	Ratio
Train	90%
Calibration	2%
Validation	4%
Test	4%

3.5 Obstacle Generation via Superellipses

We generate obstacles using superellipses (Lamé curves), which provide a flexible family of shapes:

$$\left(\frac{|x - c_x|}{a}\right)^n + \left(\frac{|y - c_y|}{b}\right)^n \leq 1 \tag{1}$$

where (c_x, c_y) is the center, (a, b) are semi-axes, and n is the shape exponent:

- $n = 1$: Diamond/rhombus
- $n = 2$: Ellipse
- $n > 2$: Squircle (approaches rectangle as $n \to \infty$)

We support rotation by angle θ for orientation diversity. The exponent n is sampled from configurable distributions:

Table 4. Superellipse exponent sampling distributions.

Distribution	Properties	Use Case
Log-uniform	Favors smaller n	Mixed shapes (default)
Uniform	Equal probability	Balanced variety
Beta	Configurable skew	Controlled bias
Exponential	Rapid decay	Primarily round shapes

The generation algorithm (Algorithm 1) iteratively places non-overlapping superellipses until reaching a target coverage ratio (sampled uniformly from $[0, \text{ratio_max}]$).

In the repository defaults, obstacle sizes are sampled as fractions of the image width/height (min/max size ratios 0.02/0.25), and generation is tuned by a fixed number of attempts (50) and candidate shapes per attempt (200). Marker positions are protected by rejecting candidates that overlap start or goal pixels.

Algorithm 1 Obstacle Generation

Require: Resolution R, config C with ratio_max, shape params
mask $\leftarrow \mathbf{0}_{R\times R}$
target_ratio $\leftarrow$ Uniform$(0, C.\text{ratio_max})$
for attempt $= 1$ to C.max_attempts **do**
  candidate $\leftarrow$ copy(mask)
  for $i = 1$ to C.candidates_per_mask **do**
    Sample $n, a, b, \theta, c_x, c_y$ from distributions
    new_mask $\leftarrow$ SuperellipseMask(R, c, a, b, n, θ)
    if coverage(candidate $\vee$ new_mask) $\leq C$.ratio_max **then**
      candidate $\leftarrow$ candidate $\vee$ new_mask
    end if
  end for
  if coverage(candidate) $\geq$ target_ratio **then**
    return candidate
  end if
end for
return mask

3.6 Obstacle Enforcement

To ensure obstacles are respected by all cost models and solvers, we apply an *obstacle penalty* during pathfinding. For cells marked as obstacles, we set:

$$e_{\text{masked}}(i, j) = \max(e) + \text{obstacle_penalty} \tag{2}$$

where the default penalty is 10^8. This approach avoids special-casing individual solvers while ensuring paths avoid obstacles across all cost models.

3.7 Cost Models

We provide five physically-motivated cost models for edge weights, enabling diverse optimization objectives:

Table 5. Cost models in Passage.

Model	Formula	Properties
Elevation	$w = d \cdot (1 + \alpha \cdot \bar{e}_{\text{dst}})$	Penalizes high destinations
Distance	$w = d$	Ignores terrain
Slope-uphill	$w = d \cdot (1 + \alpha \cdot \max(0, \Delta e)/\Delta e_{\text{range}})$	Penalizes climbing
Energy	$w = d + \alpha \cdot \max(0, \Delta e)$	Physical energy model
Absolute	$w = d \cdot (1 + \alpha \cdot \lvert\bar{e}_{\text{dst}}\rvert)$	Penalizes extremes

Where:

- d: Physical distance (30m cardinal, $30\sqrt{2} \approx 42.43$m diagonal)
- α: Configurable weight parameter (default 100.0)
- $\bar{e}$: Normalized elevation $\in [0, 1]$
- Δe: Elevation difference (destination $-$ source)

In the repository defaults, the grid backend uses the energy model with $\alpha = 100$ and Euclidean heuristic.

3.8 Data Representation

Each sample consists of:

Tensor Format. A $(H, W, 3)$ float32 NumPy array:

- Channel 0: Normalized elevation $\in [0, 1]$
- Channel 1: Marker channel (background $= -1$, start $= 0$, goal $= 1$)
- Channel 2: Obstacle mask (0=free, 1=blocked)

Metadata. A YAML file with complete provenance:

- Sample ID, resolution, split, and generation timestamp
- Source tile identifiers (center, used, overlapping) and crop geo-coordinates
- Global elevation min/max used for normalization
- Start/goal marker positions and elevations
- Obstacle coverage target/actual ratios and penalty
- Path lengths, per-pixel elevations, and cost values (with/without obstacles)
- Solver configuration (backend, name, cost model, weight, heuristic)
- Timing breakdowns (tile loading, graph prebuild, obstacle generation, pathfinding, I/O)

Tensors are stored on disk as zstd-compressed NumPy blobs (`.npy.zst`) for compactness and loaded losslessly during export and analysis.

3.9 Deterministic Sampling Strategy

For each sample, we draw a target elevation, select a tile whose min/max spans that elevation, and load a local grid of neighboring tiles to ensure full-coverage crops at large resolutions. We then select a random crop that contains the target-elevation pixel and place markers by fixing the start at the target pixel and sampling a distinct end pixel. This procedure guarantees that each crop includes a representative elevation target while maintaining stochastic diversity.

4 Generation Pipeline

4.1 Pipeline Overview

The PASSAGE toolkit provides a complete pipeline:

1. **Download**: Fetch ALOS AW3D30 tiles from JAXA servers
2. **Calibrate**: Compute global elevation statistics across all tiles
3. **Generate**: Sample crops, place markers and obstacles, compute paths
4. **Export**: Push datasets to Hugging Face Hub with zstd compression

4.2 Download Stage

The download module constructs URLs for JAXA archive tiles, tracks status in CSV format, and supports resume for interrupted downloads. Archive extraction selectively extracts DSM GeoTIFF files while discarding auxiliary data.

4.3 Calibration Stage

Calibration scans all downloaded tiles in parallel to compute:

- Global minimum and maximum elevation
- Per-tile statistics (min, max, mean, standard deviation)

These statistics enable consistent normalization across all samples and stratified sampling by elevation bands.

4.4 Sample Generation

For each target resolution R:

1. Draw random target elevation from [global_min, global_max]
2. Filter tiles whose min/max spans the target elevation; sample a center tile
3. Load a grid of neighboring tiles (handling $\pm 180°$ wrapping) and find the closest pixel to the target elevation within the center tile
4. Sample a random $R \times R$ crop that contains the target pixel
5. Place markers: start at the target pixel, goal at a distinct random pixel with a valid path
6. Generate obstacle mask via Algorithm 1 (with marker protection)
7. Compute ground-truth paths (free terrain and with obstacles) and path costs
8. Save tensor, metadata (including timing), and optional PNG visualization

4.5 Export to Hugging Face Hub

The export module supports direct push to Hugging Face Hub:

- Zstandard (zstd) compression at configurable levels
- Per-resolution chunk sizes controlling shard granularity
- Token-based authentication via `.env.secret`
- Retry/resume semantics for robustness
- Automatic dataset card generation

We provide `scripts/estimate_chunk_sizes.py` to recommend chunk sizes for target shard file sizes.

5 Solvers and Evaluation Metrics

5.1 Solver Backends

Table 6. Solver backends in Passage.

Backend	Library	Features	Best For
NetworkX	`networkx`	All algorithms, all cost models	Flexibility
Pathfinding	`pathfinding`	Grid-optimized, limited cost models	Small grids
Grid	NumPy/Numba	JIT-compiled, all cost models	Large grids

The grid backend uses NumPy with optional Numba JIT compilation when available; the NetworkX backend reuses cached graph templates and updates edge weights per crop to reduce overhead.

5.2 Solver Algorithms

Table 7. Solver algorithms.

Algorithm	Backend(s)	Complexity	Notes
A*	All	$O(E \log V)$	Heuristic-guided, optimal
A* Bidirectional	NetworkX	$O(E \log V)$	Faster for long paths
Dijkstra	All	$O(E \log V)$	No heuristic, optimal
Dijkstra Bidirectional	NetworkX	$O(E \log V)$	Symmetric search
Bellman-Ford	NetworkX	$O(VE)$	Handles negative weights
Best-First	Pathfinding	$O(E \log V)$	Greedy, not optimal

5.3 Heuristic Functions

For A* and Best-First search, we provide three admissible heuristics:

$$h_{\text{euclidean}}(s, g) = 30 \cdot \sqrt{(i_g - i_s)^2 + (j_g - j_s)^2} \tag{3}$$

$$h_{\text{manhattan}}(s, g) = 30 \cdot (|i_g - i_s| + |j_g - j_s|) \tag{4}$$

$$h_{\text{chebyshev}}(s, g) = 30 \cdot \max(|i_g - i_s|, |j_g - j_s|) \tag{5}$$

Euclidean is recommended for 8-connectivity; Manhattan for 4-connectivity.

5.4 Evaluation Metrics

Path Quality Metrics.

- **Path length (pixels)**: Number of cells in path
- **Path length (meters)**: $\sum_i d_i$ physical distance
- **Path cost**: $\sum_i w(e_i)$ under chosen cost model
- **Optimality gap**: $(c_{\text{pred}} - c^*)/c^*$
- **Success rate**: Fraction of samples with valid paths

Computational Metrics.

- **Solve time**: Wall-clock time for path computation
- **Graph build time**: Time to construct graph structure
- **Memory usage**: Peak memory during computation
- **Scalability factor**: $t(2n)/t(n)$ for resolution doubling

Learning Metrics.

- **Pixel accuracy**: Correct path pixels / total path pixels
- **IoU**: $|P_{\text{pred}} \cap P_{\text{true}}|/|P_{\text{pred}} \cup P_{\text{true}}|$
- **MSE**: Mean squared error for cost-to-go prediction
- **Hausdorff distance**: Maximum path shape deviation

Path Divergence Metrics. For comparing two paths as ordered point sequences P and Q, we also provide notebook implementations for:

- **Normalized Hausdorff distance**: $H(P, Q)/D$ where D is the straight-line marker distance
- **Chamfer (mean deviation)**: symmetric average nearest-neighbor distance normalized by D
- **Discrete Fréchet distance**: ordering-aware deviation, normalized by D
- **DTW distance**: Dynamic Time Warping normalized by $\max(|P|, |Q|)\, D$
- **Area between paths**: filled polygon area normalized by D^2

Certification Metrics.

- **Worst-case error**: $\max_i(\text{error}_i)$
- **OOD detection rate**: Detected OOD samples / total OOD
- **Calibration error**: $\mathbb{E}[|P(Y|f(X) = p) - p|]$
- **Coverage guarantee**: $P(\text{true} \in \text{interval})$

6 Experiments

6.1 Solver Timing Comparison

We benchmark the three solver backends across resolutions using A* with the elevation cost model. Figure 1 shows median solve times.

Figure Placeholder: Solver Timing Comparison

Log-log plot with:
- X-axis: Resolution (64 to 4096)
- Y-axis: Median solve time (ms)
- Three lines: NetworkX (blue), Pathfinding (orange), Grid (green)
- Expected trend: Grid backend $\sim$10$\times$ faster at 4096$\times$4096

Figure 1. Solver computation time across resolutions (log-log scale). The NumPy-grid backend with Numba JIT compilation achieves the best scaling for resolutions ≥ 1024, while NetworkX provides comprehensive algorithm support with higher per-instance overhead.

Key findings:

- NetworkX provides full feature support but higher per-instance overhead
- Pathfinding library is efficient for small grids but lacks directional cost models
- NumPy-grid with Numba JIT compilation achieves best scaling for ≥ 1024 resolution

6.2 Obstacle Coverage Distribution

Figure 2 shows the distribution of obstacle coverage ratios across generated samples, and Figure 3 presents example samples at different resolutions.

Figure Placeholder: Obstacle Coverage Distribution

Histogram showing:
- X-axis: Obstacle coverage ratio (0.0 to 0.30)
- Y-axis: Number of samples
- Near-uniform distribution due to uniform target sampling
- Slight right-skew from non-overlap constraints

Figure 2. Distribution of obstacle coverage ratios across PASSAGE samples. Target coverage is sampled uniformly from $[0, 0.30]$; the slight right-skew results from non-overlap constraints occasionally preventing target coverage from being reached.

6.3 Cost Model Sensitivity

We provide a dedicated notebook to compare all cost models under a fixed solver (grid A*) and multiple weight settings. The analysis highlights how the same terrain induces different path morphologies when optimizing elevation, slope-uphill, energy, or pure distance. This study is useful for selecting cost models that best match a given operational objective.

Figure Placeholder: Sample Gallery

2×4 grid showing samples at resolutions 64, 128, 256, 512, 1024, 2048, 4096:

Each panel shows:
• Elevation map (grayscale, darker=lower)
• Obstacles (red overlay)
• Start marker (green circle)
• Goal marker (blue circle)
• Ground-truth path (yellow line)

Demonstrates increasing terrain detail and path complexity with resolution.

Figure 3. Example samples from PASSAGE at different resolutions. Each panel shows normalized elevation (grayscale), superellipse obstacles (red), start/goal markers (green/blue), and the ground-truth path (yellow). Higher resolutions capture finer terrain detail and enable more complex path geometries.

6.4 Path Divergence Metrics

Another notebook evaluates path divergence metrics (Hausdorff, Fréchet, DTW, Chamfer, and area-based measures) across pairs of elevation crops with identical markers. Metrics are normalized by marker distance to reduce resolution sensitivity. This provides a quantitative lens on how terrain variability affects path stability.

6.5 Dataset Insights

The dataset-insights notebook computes global statistics from stored metadata without recomputing paths. It produces a path occurrence heatmap, tile coverage distribution, elevation histograms (start points and all path pixels), path length distributions, normalized relative elevation profiles, normalized Hausdorff distances (clean vs straight, obstacle vs clean), and identical path counts. These analyses serve as sanity checks and provide high-level dataset diagnostics.

6.6 Baseline Learning Models

We evaluate three baseline architectures on PASSAGE:

U-Net for Path Segmentation. A standard U-Net Ronneberger et al. (2015) trained to predict path masks. Input: 3-channel tensor (elevation, markers, obstacles). Output: binary path mask. Loss: weighted cross-entropy.

GNN for Node Classification. We convert elevation grids to PyTorch Geometric graphs and train a 4-layer GCN Kipf and Welling (2017) for node-level path prediction. Edge weights encode cost model values.

PPO for Navigation. A Proximal Policy Optimization Schulman et al. (2017) agent learns navigation policies on 64×64 and 128×128 grids. Reward: negative step cost under energy model. Episode terminates on goal reach or collision.

Note: Full experimental details and additional resolutions in Appendix C.

Table 8. Baseline results on PASSAGE test set (256×256 resolution).

Model	Pixel Acc.	IoU	Success %
U-Net	0.87	0.71	94.2
GNN (GCN)	0.82	0.65	89.7
PPO (64×64)	—	—	78.3

7 Proposed Research Directions

PASSAGE enables several research directions:

7.1 Generalization and Distribution Shift

- **Resolution transfer**: Train on small resolutions, evaluate on large
- **Elevation extrapolation**: Train on mid-elevation, test on extremes
- **Obstacle density shift**: Train on low coverage, test on high

7.2 Certification-Ready Models

- **Conformal prediction**: Use calibration split to construct prediction intervals with coverage guarantees
- **Uncertainty quantification**: Bayesian neural networks, ensembles, evidential deep learning
- **Formal verification**: Lipschitz bounds, certified robustness to perturbations

7.3 Scalable Architectures

- **Efficient transformers**: Linear attention, sparse patterns for high-resolution inputs
- **Neural algorithmic reasoning**: Differentiable dynamic programming, pointer networks
- **Hierarchical planning**: Multi-scale approaches combining coarse and fine paths

7.4 Future Dataset Extensions

- **Ordered waypoints**: Intermediate points that must be visited in sequence
- **Multi-agent coordination**: Collision-free paths for multiple agents
- **Temporal dynamics**: Time-varying obstacles, scheduling constraints
- **Flight model constraints**: Turn radius, climb/descent rates, energy budgets

8 Limitations

We acknowledge several limitations of the current PASSAGE release:

Static Environments. All samples represent static snapshots—obstacles do not move, and terrain does not change over time. Real-world applications may require handling dynamic obstacles (*e.g.*, weather, temporary no-fly zones) and re-planning capabilities.

Single-Agent Focus. The dataset currently supports only single-agent pathfinding. Multi-agent path finding (MAPF) scenarios Stern et al. (2019) with collision avoidance constraints are deferred to future work.

Simplified Cost Models. While we provide five cost models, they remain simplified approximations of real-world traversal costs. Factors such as wind, vehicle dynamics, fuel consumption curves, and sensor constraints are not modeled.

Resolution-Sample Tradeoff. Due to computational constraints, higher resolutions have fewer samples (only 5 samples at 4096×4096). This may limit statistical power for evaluating model performance at extreme resolutions.

Geographic Bias. While ALOS AW3D30 provides near-global coverage, the random sampling procedure may not uniformly represent all terrain types (*e.g.*, deserts, mountains, coastal areas). Stratified sampling by terrain type could improve diversity.

Obstacle Realism. Superellipse obstacles provide geometric variety but do not represent real-world obstacle patterns (*e.g.*, buildings, vegetation, restricted airspace). Domain-specific obstacle models may be needed for certain applications.

9 Broader Impact

Positive Applications. PASSAGE is designed to advance research in certifiable machine learning for safety-critical navigation systems. Potential beneficial applications include:

- **Search and rescue**: Optimizing UAV routes for locating missing persons in mountainous terrain
- **Disaster response**: Planning efficient paths for emergency supply delivery
- **Environmental monitoring**: Routing autonomous vehicles for wildlife surveys with minimal terrain disturbance
- **Accessibility**: Developing terrain-aware routing for mobility-impaired individuals

Potential Risks. As with any navigation technology, there are dual-use concerns:

- **Military applications**: Pathfinding algorithms could be adapted for autonomous weapons systems. We encourage users to consider ethical implications and adhere to responsible AI principles.
- **Privacy**: While PASSAGE uses publicly available elevation data with 30m resolution (insufficient for identifying individuals or structures), derived applications should consider privacy implications of high-resolution terrain analysis.
- **Environmental impact**: Optimized routing could increase access to previously inaccessible areas, potentially enabling environmental degradation. Applications should incorporate ecological constraints.

Data Licensing. The underlying ALOS AW3D30 data is provided by JAXA under terms permitting research use. Users should review JAXA's data policy for commercial applications. The PASSAGE toolkit and derived dataset are released under permissive open-source licenses to maximize research accessibility.

Carbon Footprint. Full dataset generation requires significant computation. We provide pre-generated datasets on Hugging Face Hub to reduce redundant computation. Users are encouraged to reuse existing splits rather than regenerating.

10 Conclusion

We introduced PASSAGE, a large-scale, multi-resolution elevation-based pathfinding dataset designed for certifiable machine learning systems. Key features include:

- **Scale**: 141,300 samples across seven resolutions (64×64 to 4096×4096)
- **Realism**: Continuous elevation data from JAXA ALOS AW3D30
- **Flexibility**: Five cost models, three solver backends, configurable superellipse obstacles
- **Certification alignment**: Bounded ODD, provenance metadata, calibration split

- **Accessibility**: Direct Hugging Face Hub export with zstd compression

PASSAGE enables rigorous benchmarking of supervised surrogate models, reinforcement learning agents, and graph neural networks while providing the structure needed for certification-oriented evaluation. Future work will extend the dataset with waypoints, multi-agent scenarios, and flight-model constraints.

Availability. Dataset and code: https://huggingface.co/datasets/passage

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A Configuration Schema

The full configuration schema (`config/config.yaml`):

Listing 1. Configuration schema excerpt

```yaml
resolutions: [64, 128, 256, 512, 1024, 2048, 4096]

num_samples:
    64: 100000
    128: 25000
    256: 10000
    512: 5000
    1024: 1000
    2048: 200
    4096: 100

splits:
    enabled: true
    ratios:
        train: 0.9
        calibration: 0.02
        validation: 0.04
        test: 0.04
    seed: 42

solver:
    name: grid:astar
    grid:
        weight: 100.0
        cost_model: energy
        heuristic: euclidean

obstacles:
    enabled: true
    ratio_max: 0.30
    shape:
        n:
            min: 1.0
            max: 80.0
            distribution:
                type: log_uniform
```

```
    obstacle_penalty: 1e8

export:
    chunk_size:
        64: 107820
        128: 39079
        256: 12133
        512: 4425
        1024: 909
        2048: 280
        4096: 83
```

B CLI Reference

Listing 2. CLI commands

```
# Download ALOS tiles
passage download --outputs-dirpath outputs/download --num-cpus 8

# Calibrate elevation ranges
passage calibrate --tiles-dirpath outputs/download/tiles

# Generate dataset
passage generate --resolution 64 128 256 --num-samples 1000

# Export to Hugging Face
passage export --repo-id username/passage-dataset
```

C Extended Experiments

Extended baseline results across all resolutions and ablation studies on cost models are provided in the supplementary materials accompanying this submission.

D Reproducibility Checklist

- ✓ Code released under open-source license
- ✓ Dataset available on Hugging Face Hub
- ✓ Configuration files included
- ✓ Random seeds specified
- ✓ Hardware requirements documented
- ✓ Provenance metadata per sample